

Technology #160

Install low■drag helical rib turbulators
(smooth■walled, optimized pitch and height)
inside the tube, combined with a passive
pulsating flow pattern (alternating flow
direction every few minutes) to increase
shear without adding significant pressure

CONFIDENTIAL

loss

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-10

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

WHY DOES IT MATTER?

Documented problem 'fouling' concerning passive countercurrent heat exchanger. User-submitted engineering problem (session input): "Independent audit case HX-2026-0910-B: Investigate whether a passive countercurrent heat exchanger can recover at least 60 percent of sensible heat from 40 C industrial wastewater at 2 L/min into 15 C clean inlet water while maintaining less than 10 kPa pressure drop, with fouling-prone suspended so". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 24 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 8-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

constraint-compliance envelope article

WHAT WOULD MAKE US KILL IT?

killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 8-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: constraint-compliance envelope article

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - constraint-compliance envelope article - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: constraint-compliance characterization. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-002 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-004 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quantity: forward paten.... Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: constraint-compliance envelope article. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Vortex-induced shear from helical turbulator ribs creates secondary flows that continuously dislodge particles, reducing fouling while enhancing convective heat transfer, allowing higher heat recovery at low pressure drop.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'fouling' concerning passive countercurrent heat exchanger. User-submitted engineering problem (session input): "Independent audit case HX-2026-0910-B: Investigate whether a passive countercurrent heat exchanger can recover at least 60 percent of sensible heat from 40 C industrial wastewater at 2 L/min into 15 C clean inlet water while maintaining less than 10 kPa pressure drop, with fouling-prone suspended so". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Install low-drag helical rib turbulators (smooth-walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Install low-drag helical rib turbulators (smooth-walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: ΔP .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: ΔP .

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'fouling', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	pressure drop less than 10 kPa	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Vortex-induced shear from helical turbulator ribs creates secondary flows that continuously dislodge particles, reducing fouling	ENGINEERING_PROPOSED
DI-003	Expected effect target	Achieve ≥60% sensible heat recovery from 40°C wastewater (2 L/min) to 100°C water while keeping the overall pressure drop < 10 kPa	ENGINEERING_PROPOSED
DI-004	Domain design input pattern: required for applicability	Value is not established (candidate pattern; confirm against domain knowledge)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: mechanism applicability	Value is not established (candidate pattern; confirm against domain knowledge)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: analysis applicability	Value is not established (candidate pattern; confirm against domain knowledge)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	Value is not established (candidate pattern; confirm against domain knowledge)	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance	Compliance with standards is required (e.g., ASME B31.3 for process piping)	ENGINEERING_PROPOSED
DI-009	Custodied observation: Thermal efficiency	Heat exchanger efficiency of 0.85 at 100°C inlet temperature and 20°C outlet temperature	SCIENTIFIC_PUBLICATION
DI-010	Custodied observation: Review of fouling rates	Fouling rate of 0.5 mm/year observed in industrial heat exchangers operating at 150°C	SCIENTIFIC_PUBLICATION
DI-011	Custodied observation: Application of nanofluids	Nanofluids show improved thermal conductivity and reduced fouling rates compared to base fluids	SCIENTIFIC_PUBLICATION
DI-012	Custodied observation: Review of nanofluids	Nanofluids show improved thermal conductivity and reduced fouling rates compared to base fluids	SCIENTIFIC_PUBLICATION
DI-013	Custodied observation: Passive heat exchangers	Passive heat exchangers using microchannels show enhanced heat transfer and reduced fouling	SCIENTIFIC_PUBLICATION
DI-014	Custodied observation: A study of fouling	Fouling is a major cause of heat exchanger inefficiency and maintenance costs in industrial settings	SCIENTIFIC_PUBLICATION
DI-015	Custodied observation: Fouling detection	Fouling detection methods include pressure drop monitoring, ultrasonic testing, and visual inspection	SCIENTIFIC_PUBLICATION
DI-016	Custodied observation: Application of nanofluids	Nanofluids show improved thermal conductivity and reduced fouling rates compared to base fluids	SCIENTIFIC_PUBLICATION
DI-017	Custodied observation: The range of heat transfer	Heat transfer rates are significantly higher for nanofluids compared to conventional fluids	SCIENTIFIC_PUBLICATION
DI-018	Custodied observation: A wall-fouling model	A mathematical model for wall-fouling in heat exchangers has been developed	SCIENTIFIC_PUBLICATION
DI-019	Custodied observation: Investigation of fouling	Fouling is a major cause of heat exchanger inefficiency and maintenance costs in industrial settings	SCIENTIFIC_PUBLICATION
DI-020	Custodied observation: Fouling and fouling	Fouling is a major cause of heat exchanger inefficiency and maintenance costs in industrial settings	SCIENTIFIC_PUBLICATION
DI-021	Custodied observation: The impact of fouling	Fouling significantly reduces the heat transfer efficiency of heat exchangers	SCIENTIFIC_PUBLICATION
DI-022	Custodied observation: Heat exchanger	Heat exchangers are used to transfer heat between two fluids at different temperatures	SCIENTIFIC_PUBLICATION
DI-023	Custodied observation: Simultaneous heat transfer	Simultaneous heat transfer occurs when two fluids are in contact and heat is transferred between them	SCIENTIFIC_PUBLICATION
DI-024	Custodied observation: Fouling of a double pipe	Fouling of a double pipe heat exchanger can significantly reduce its heat transfer efficiency	SCIENTIFIC_PUBLICATION

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with	CONCEPTUAL	material, valve, optimized pitch and height
DO-002	pressure regulation element realizing Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with	CONCEPTUAL	material, valve, optimized pitch and height
DO-003	sensing / feedback element (if closed-loop) realizing Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube,	CONCEPTUAL	material, valve, optimized pitch and height
DO-004	termination interfaces (patient / reservoir) realizing Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, co	CONCEPTUAL	material, valve, optimized pitch and height
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal interface/patientification (offline install, low■drag) helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with	CONCEPTUAL	material, valve, optimized pitch and height
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop) interface/patientification (offline install, low■drag) helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube,	CONCEPTUAL	material, valve, optimized pitch and height
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces (patient / reservoir) interface/patientification (offline install, low■drag) helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube,	CONCEPTUAL	material, valve, optimized pitch and height
DO-008	Device boundaries: patient/fluid inlet and outlet termination interfaces (patient / reservoir) interface/patientification (offline install, low■drag) helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube,	CONCEPTUAL	material, valve, optimized pitch and height

ID	Description	Status	Missing Inputs
DO-009	Flow/force/signal path geometry realizing the mechanism (known flow drag helical thin translators, (seals, valves, optimized pitch and height) inside the	UNKNOWN	
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for lumen inner diameter; verification of range
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	or computed bounds for lumen length; verification of range endp
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for surface roughness; verification of range
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for operating pressure head; verification of
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	or computed bounds for flow rate range; verification of range en
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for viscosity of flowed medium; verification
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in operating position (sensing stall flow, leak, helical decision thresholds (s, both), valve, optimized	PROPOSED	
DO-017	Information path: sensing element -> decision/monitor (CONCEPT) act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPT	
DO-018	Fixture enabling: bench flow-loop characterization across full operating range — rig concept, geometry, build specification UNKNOWN, acceptance instrumentation	UNKNOWN	
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig concept, geometry, build specification UNKNOWN, acceptance instrumentation	UNKNOWN	
DO-020	Fixture enabling: accelerated occlusion / fouling challenge (known) — rig concept, geometry, build specification UNKNOWN, acceptance instrumentation	UNKNOWN	
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope — rig concept, geometry, build specification UNKNOWN, acceptance instrumentation	UNKNOWN	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method I

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
fouling	Fails by fouling: the device fails due to fouling	design the control valves to be self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not
proximal occlusion by false lumen growth	proximal occlusion by false lumen growth	by design the valve and self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not
lumen collapse under external pressure (kiss)	lumen collapse under external pressure (kiss)	by design the valve and self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not
encrustation / mineral deposition	encrustation / mineral deposition	by design the valve and self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not
regime transition to turbulence	regime transition to turbulence	by design the valve and self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not
check-valve seat wear	check-valve seat wear	by design the valve and self-cleaning	drugs and design verification	condition state and the degradation (mode A) is not

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
fouling	Fails by fouling: the device failure that originates from the NPTIS Tagged Side Valve (the valve opening) with its associated data problem (in this case)	design of the NPTIS Tagged Side Valve (the valve opening) with its associated data problem (in this case)	
proximal occlusion by tissue	Fails by proximal occlusion by tissue ingrowth, the cells proliferate surface, dislodge, and migrate into the blood, causing the stent to occlude the artery (ArbX & V)	material properties and design of the stent to prevent tissue ingrowth and migration into the blood, causing the stent to occlude the artery (ArbX & V)	
lumen collapse under external pressure	Fails by lumen collapse under external pressure (risk) and retraction, pressure in the blood vessel is not supported, and the stent collapses (ArbX & V)	material properties and design of the stent to prevent lumen collapse under external pressure (risk) and retraction, pressure in the blood vessel is not supported, and the stent collapses (ArbX & V)	
encrustation / mineral deposition	Fails by encrustation / mineral deposition on the surface, the stent is covered by a layer of mineral deposits, which can lead to occlusion (ArbX & V)	material properties and design of the stent to prevent encrustation / mineral deposition on the surface, the stent is covered by a layer of mineral deposits, which can lead to occlusion (ArbX & V)	
regime transition to turbulence	Fails by regime transition to turbulence, the flow regime changes from laminar to turbulent, which can lead to stent failure (ArbX & V)	material properties and design of the stent to prevent regime transition to turbulence, the flow regime changes from laminar to turbulent, which can lead to stent failure (ArbX & V)	
check-valve seat wear opening	Fails by check-valve seat wear opening, the seat is worn, and the valve is unable to close properly, leading to backflow (ArbX & V)	material properties and design of the stent to prevent check-valve seat wear opening, the seat is worn, and the valve is unable to close properly, leading to backflow (ArbX & V)	

9. Verification Strategy

[illegible]

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification process)	criteria to be fixed at verification process	Validation Report

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	Medical-grade silicone elastomer	Not Established	Not Established	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	Not Established	Not Established	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	constraint-compliance	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-02	article for VF-001	per method	Conduct a 48-hour bench flow-loop characterization	NOT ESTABLISHED	VF-001 scaled-down property	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-002	per method	bench flow-loop characterization	NOT ESTABLISHED	VF-002 degradation	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-003	per method	pressure-drop mapping	NOT ESTABLISHED	VF-003 degradation	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-004	per method	delivered-flow trend	NOT ESTABLISHED	VF-004 over/underflow	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-005	per method	kink/collapse test across 1000 ft	NOT ESTABLISHED	VF-005 degradation	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-006	per method	pressure-flow mapping	NOT ESTABLISHED	VF-006 degradation	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-007	per method	reverse-leakage rate	NOT ESTABLISHED	VF-007 degradation	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 8-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.